

Indian Institute of Information Technology

A DESIGN OF SOFTWARE SYSTEMS(CS612) PROJECT

BiteAhead!: A Food Pre-Ordering System for Canteens

Team Members:

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PROBLEM STATEMENT

The existing manual or semi-digital canteen ordering faces several challenges:

1. Long queues and increased waiting time during peak hours.
2. Difficulty checking food availability before reaching the counter.
3. Manual order-processing errors.
4. Difficulty handling multiple simultaneous orders.
5. Poor order-status visibility.
6. Inefficient pickup-time and inventory management.

Need: A centralized digital solution to streamline food ordering, inventory, order processing and pickup.

PROPOSED SOLUTION

A centralized Canteen Pre-order System designed for convenient campus food ordering and scheduled pickup.
key features:

1. Browse menu and check food availability.
2. Add items to cart and select quantities.
3. Make or simulate payments.
4. Track orders using a unique Order ID.
5. Receive order-status notifications.
6. Dedicated interfaces for customers, staff and administrators.

Architecture: Service-oriented Architecture (SOA)with independent services communicating through REST APIs

REQUIREMENT ANALYSIS

Functional requirements

1. Authentication and user management.
2. Menu and Inventory Management.
3. Cart and pre-order management.
4. Payment and order processing.
5. Pickup Verification and Cancellation.
6. Notifications.

Non-Functional Requirements

- Performance and Reliability.
- Security and data integrity.
- Usuability and scalability.
- Maintainability and Interoperability.
- Privacy, Backup and Recovery.
- Constraints:Limited Development time, initial user capacity.

REQUIREMENT ANALYSIS

Target Users

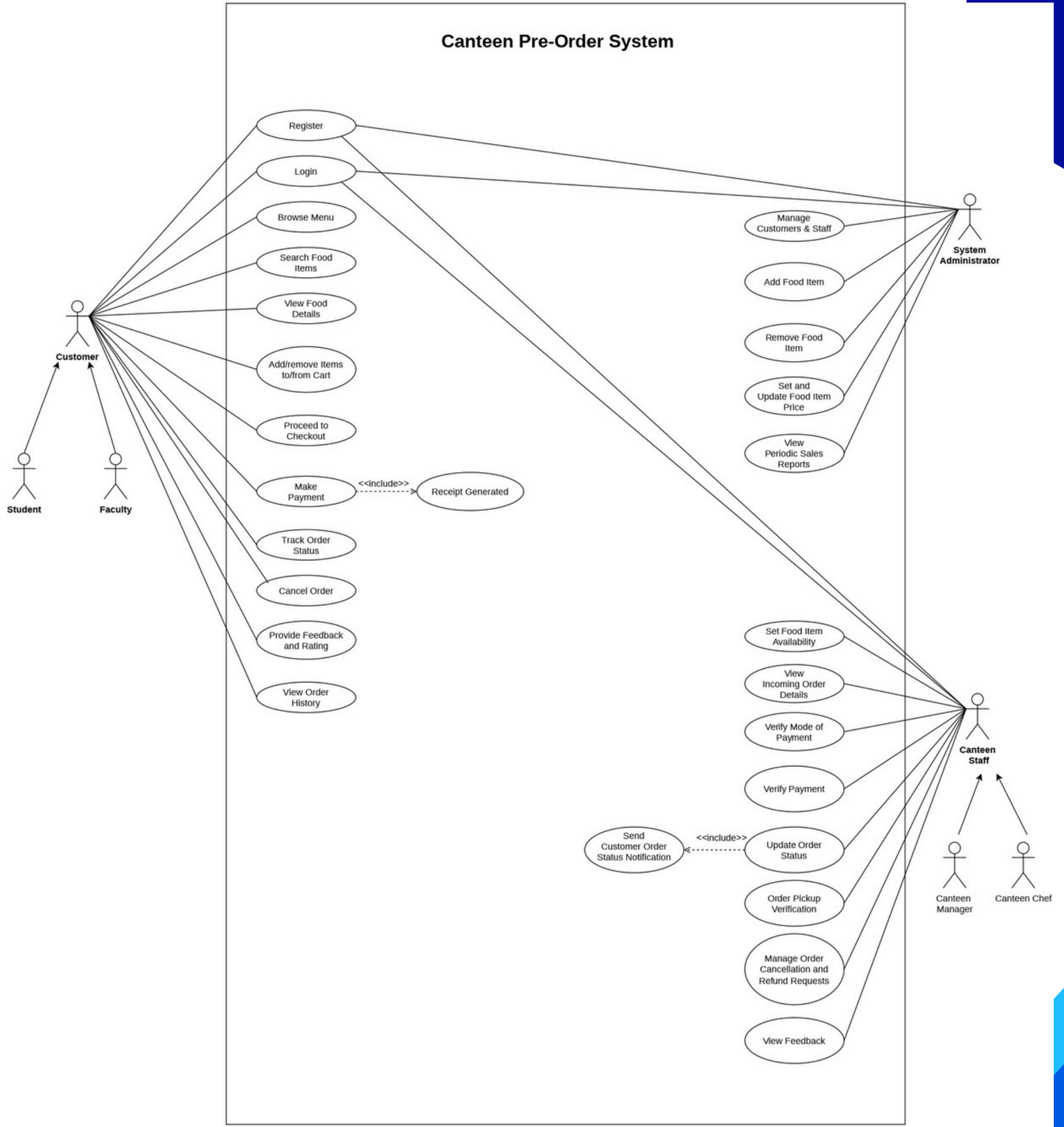
1. Student
2. Faculty
3. Canteen Staff
4. System Administrators

Technology Stack

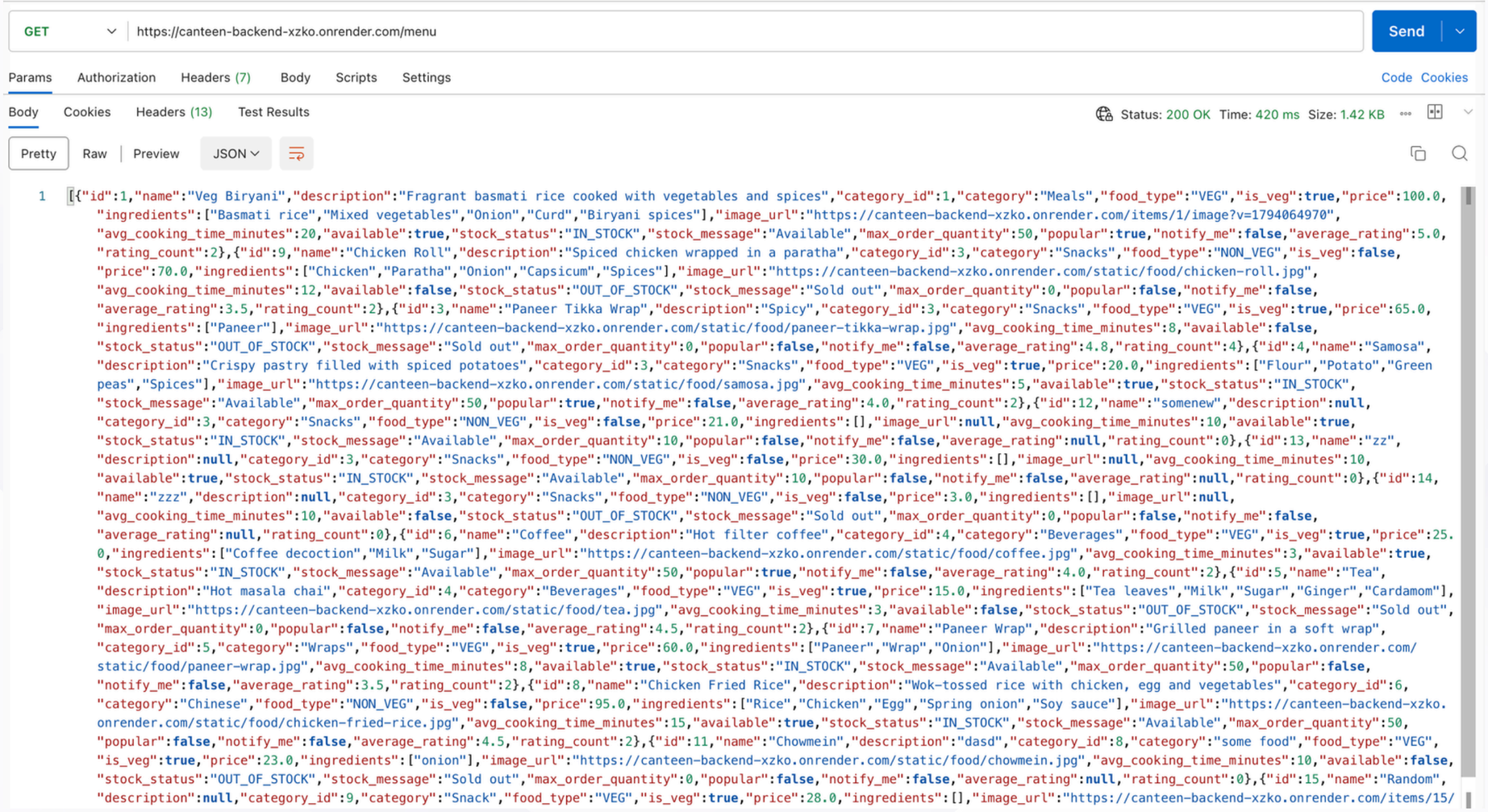
Layer	Technology	Purpose
Frontend	React Native, Javascript, CSS	User Interface
Backend	Python, FastAPI	Server-side logic and APIs
Database	PostgreSQL	Store application data
ORM	SQLAlchemy	Database interaction
Authentication	JWT	User interaction

DESIGN

Use-Case Diagram



IMPLEMENTATION: BACKEND – API Endpoint (Customers)




```
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GET https://canteen-backend-xzko.onrender.com/menu - For displaying the Menu (in Stock) for ordering

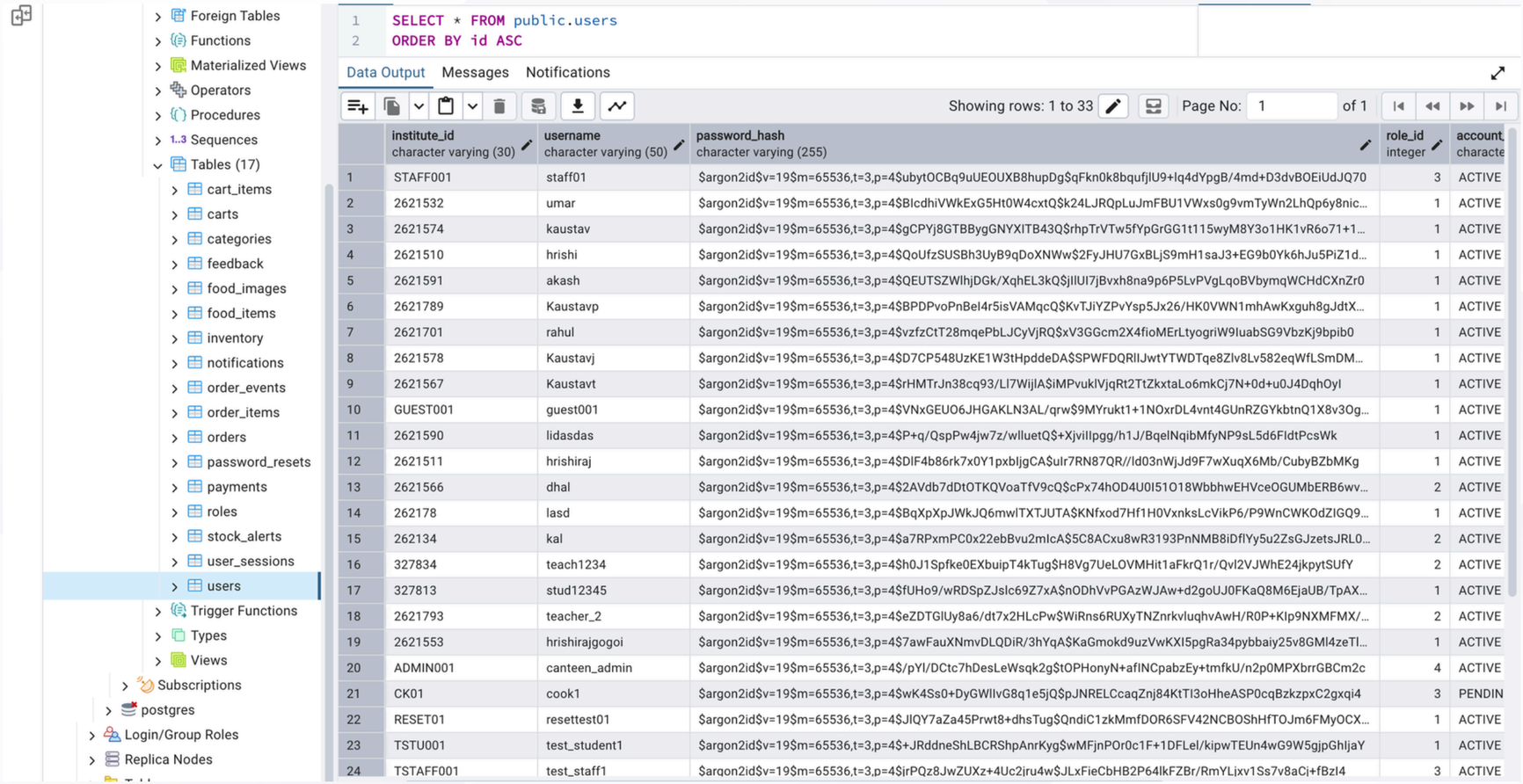
IMPLEMENTATION: BACKEND – All API Endpoints

Entity	Endpoint	Body	Response
Register	/auth/{role}/register	institute_id, username, password	user_id, role
Login	/auth/{role}/login	institute_id, password	access_token, role
Cart	/cart/items	food_id, quantity	items, total
Place Order	/orders	payment_method	order_id, pickup_code
Payment	/payments/verify	razorpay ids, signature	verified: true
Feedback	/orders/{id}/feedback	food_id, rating, description	ratings saved

IMPLEMENTATION: BACKEND – All API Endpoints (Cont.)

Entity	Endpoint	Body	Response
Food Item	/admin/items	name, price, category	id, available
Inventory	/staff/inventory/{id}	quantity	stock updated
Serve Order	/staff/orders/{id}/serve	pickup_code	status: SERVED
Live Status	/orders/{id}	none	status, ETA, timeline
Admin Live Monitor	/admin/live/feed	none	live API calls, stats
Sales Report	/admin/reports/sales	period	revenue, orders, best seller

IMPLEMENTATION: BACKEND – Database (PostgresSQL)



	institute_id character varying (30)	username character varying (50)	password_hash character varying (255)	role_id integer	account_ characte
1	STAFF001	staff01	$argon2id$v=19$m=65536,t=3,p=4$ubytOCBq9uUEOUXB8hupDg$qFkn0k8bqufjlU9+lq4dYpgB/4md+D3dvBOEiUdJQ70	3	ACTIVE
2	2621532	umar	$argon2id$v=19$m=65536,t=3,p=4$BIcdhiVWkExG5Ht0W4cxtQ$k24LJRQpLuJmFBU1VWxs0g9vmTyWn2LhQp6y8nic...	1	ACTIVE
3	2621574	kaustav	$argon2id$v=19$m=65536,t=3,p=4$gCPYj8GTBBygGNYXITB43Q$rhpTrVTw5fYpGrGG1t115wyM8Y3o1HK1vR6o71+1...	1	ACTIVE
4	2621510	hrishi	$argon2id$v=19$m=65536,t=3,p=4$QoUfzSUSBh3UyB9qDoXNWw$2FyJHU7GxBLjS9mH1saJ3+EG9b0Yk6hJu5PiZ1d...	1	ACTIVE
5	2621591	akash	$argon2id$v=19$m=65536,t=3,p=4$QEUTSZWlhjDGk/XqhEL3kQ$jIlUI7jBvxh8na9p6P5LvPVgLqoBVbymqWCHdCXnZr0	1	ACTIVE
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11	2621590	lidasdas	$argon2id$v=19$m=65536,t=3,p=4$P+q/QspPw4jw7z/wlluetQ$+XjviIlpgg/h1J/BqelNqibMfyNP9sL5d6FIdtPcsWk	1	ACTIVE
12	2621511	hrishiraj	$argon2id$v=19$m=65536,t=3,p=4$DIF4b86rk7x0Y1pxbIjgCA$uIr7RN87QR//Id03nWjJd9F7wXuqX6Mb/CubyBZbMKg	1	ACTIVE
13	2621566	dhal	$argon2id$v=19$m=65536,t=3,p=4$2AVdb7dDtOTKQVoaTfV9cQ$cPx74hOD4U0I51O18WbbhwEHVceOGUMbERB6wv...	2	ACTIVE
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16	327834	teach1234	$argon2id$v=19$m=65536,t=3,p=4$h0J1Spfke0EXbuipT4kTug$H8Vg7UeLOVMHit1aFkrQ1r/Qvl2VJWhE24jkpytSUfY	2	ACTIVE
17	327813	stud12345	$argon2id$v=19$m=65536,t=3,p=4$fUHo9/wRDSpZJsIc69Z7xA$nODhVvPGAzWJAw+d2goUJ0FKaQ8M6EjaUB/TpAX...	1	ACTIVE
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19	2621553	hrishirajgogoi	$argon2id$v=19$m=65536,t=3,p=4$7awFauXNmvDLQDiR/3hYqA$KaGmokd9uzVwKXI5pgRa34pybbaiy25v8GMl4zeTl...	1	ACTIVE
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24	TSTAFF001	test_staff1	$argon2id$v=19$m=65536,t=3,p=4$irPQz8JwZUXz+4Uc2iru4w$JLxFieCbHB2P64lkFZBr/RmYLjxv1Ss7v8aCj+fBzI4	3	ACTIVE

Users/Customers Table

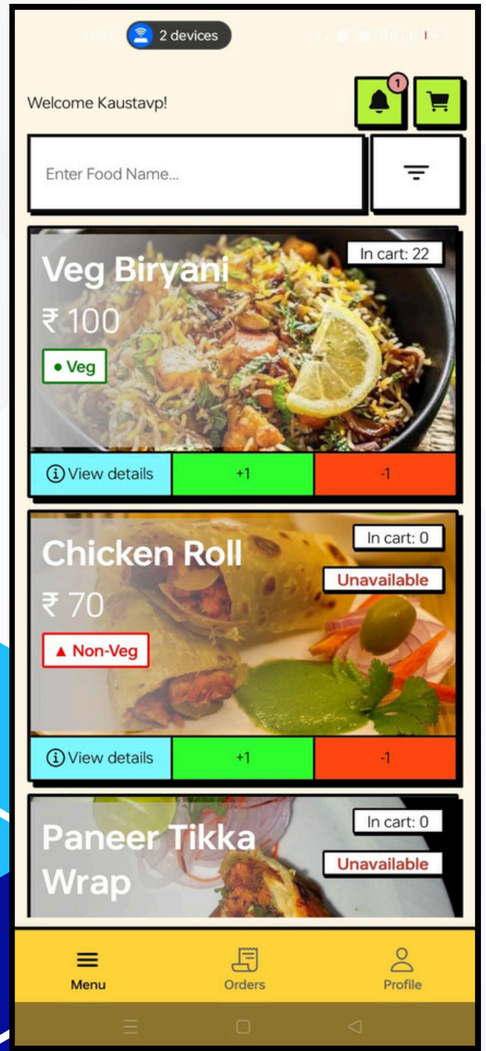
IMPLEMENTATION: BACKEND – All Tables in Database

Table	Key Fields	Purpose
roles	name	Student, Faculty, Staff, Admin
users	institute_id, username, password_hash, email, role_id, account_status	All user accounts
password_resets	user_id, code_hash, expires_at, attempts	Forgot-password OTP
food_items	name, price, category_id, food_type, ingredients, avg_cooking_time, image_url	Menu items
food_images	food_item_id, content_type, data	Uploaded food photos
inventory	food_item_id, quantity, availability	Stock per item
cart_items	cart_id, food_item_id, quantity, reserved_until	Items in cart (stock held)
orders	order_id, user_id, total_amount, status, payment_method, payment_status, pickup_code	Placed orders
order_events	order_id, event, changed_by_id	Live status timeline
payments	razorpay_order_id, razorpay_payment_id, amount, status, refund_id	Razorpay payments
feedback	order_item_id, food_item_id, rating, description	Ratings & reviews
notifications	user_id, type, message, order_id, read_at	In-app alerts

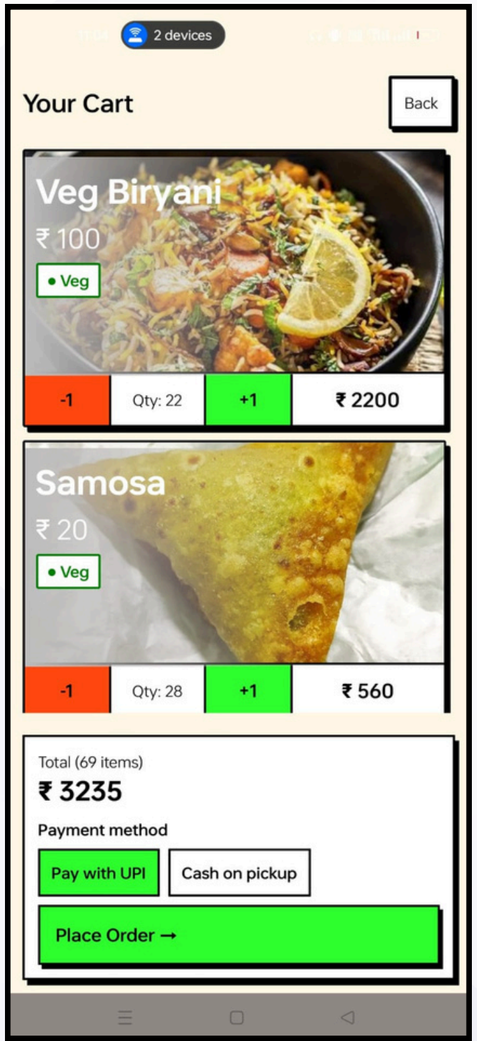
IMPLEMENTATION: Frontend

A generic success flow:

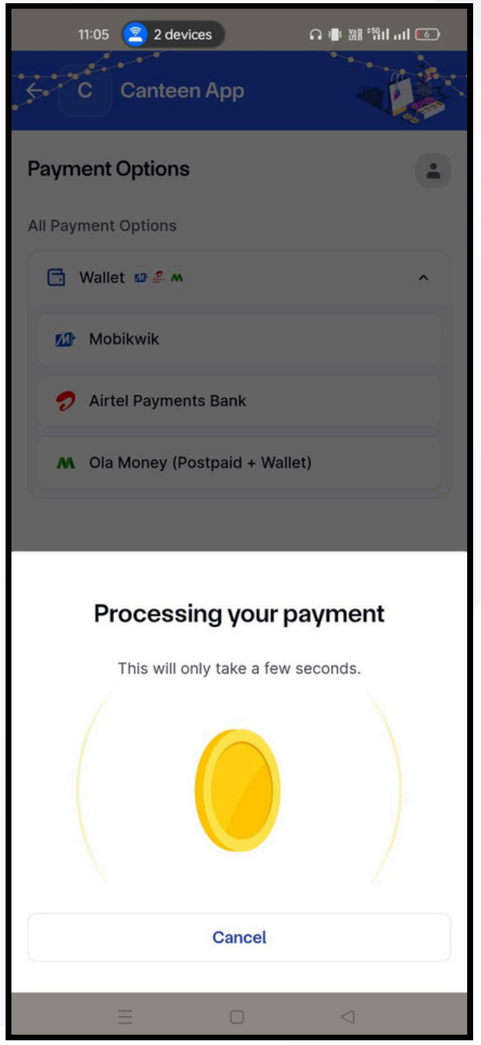
[CUSTOMER]
Browse Menu and add Items to Cart



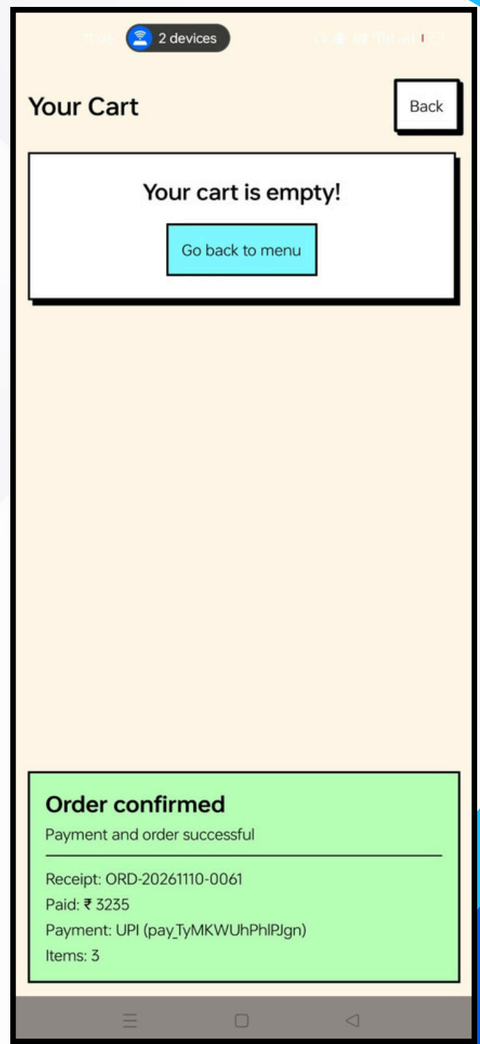
[CUSTOMER]
Cart



[CUSTOMER]
Payments



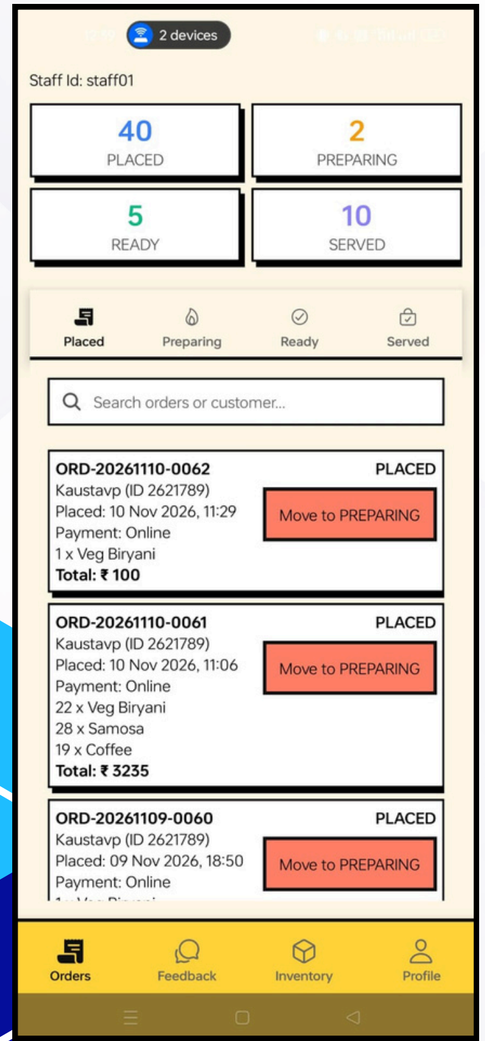
[CUSTOMER]
On Order Placed



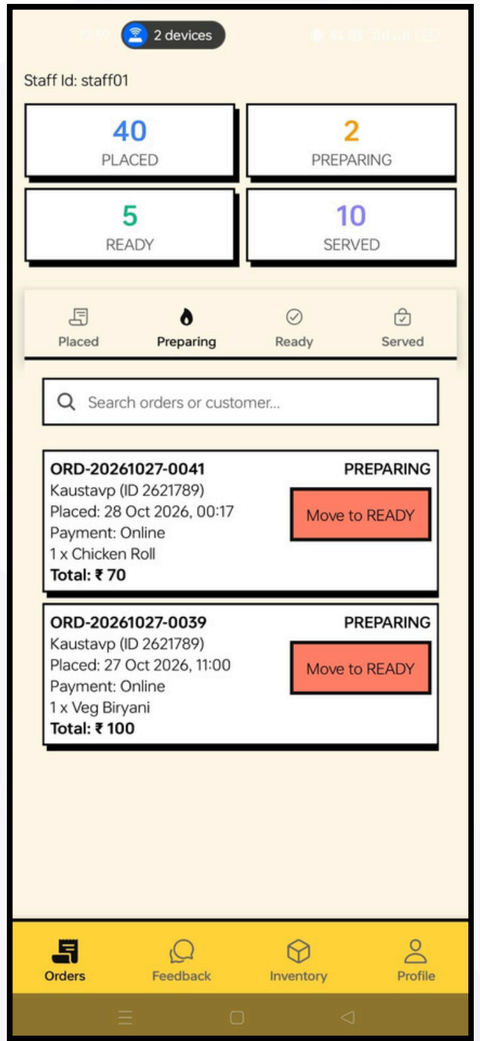
IMPLEMENTATION: Frontend

A generic success flow:

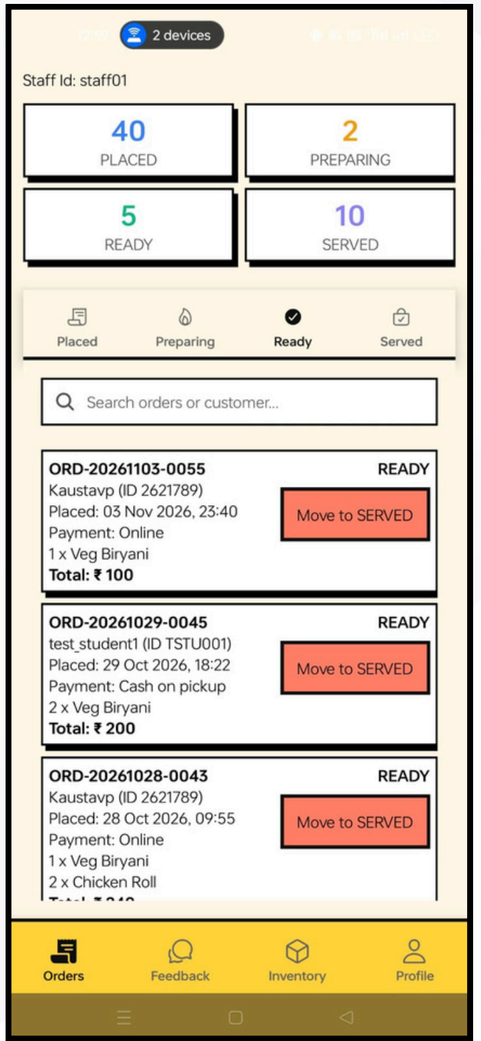
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Placed



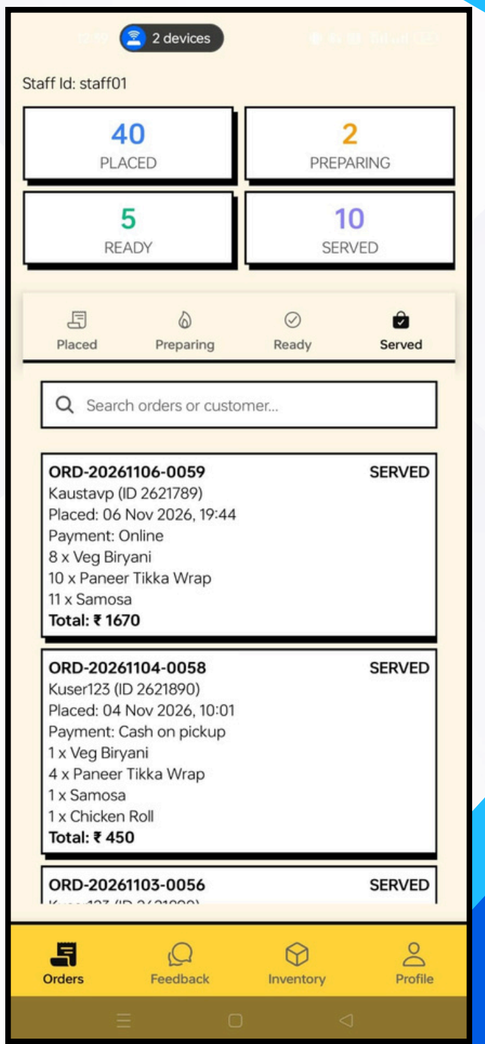
[STAFF]
Preparing



[STAFF]
Ready



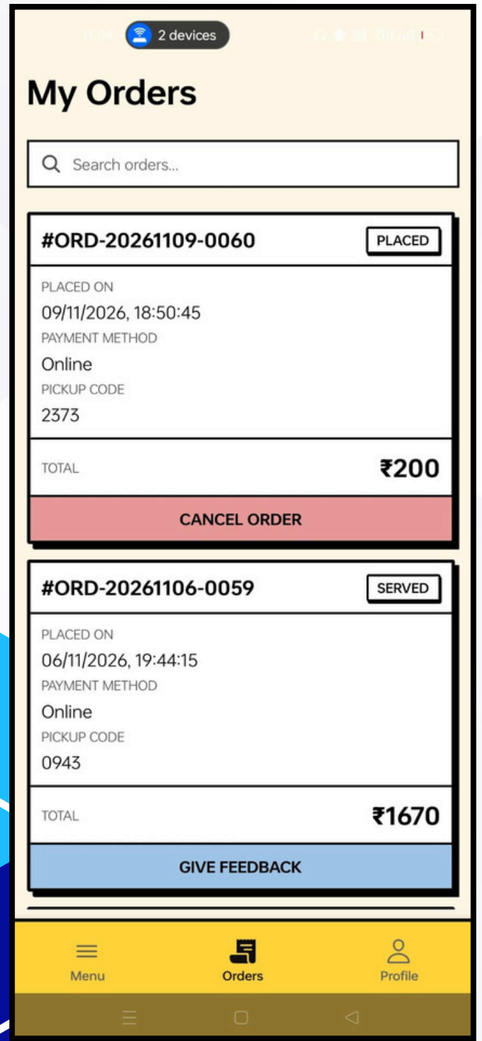
[STAFF]
Served
(After verification)



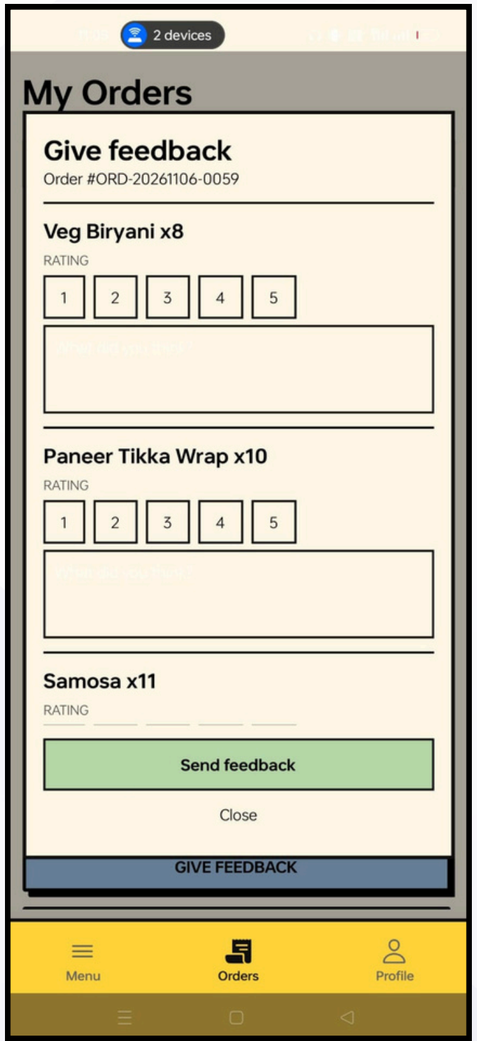
IMPLEMENTATION: Frontend

A generic success flow:

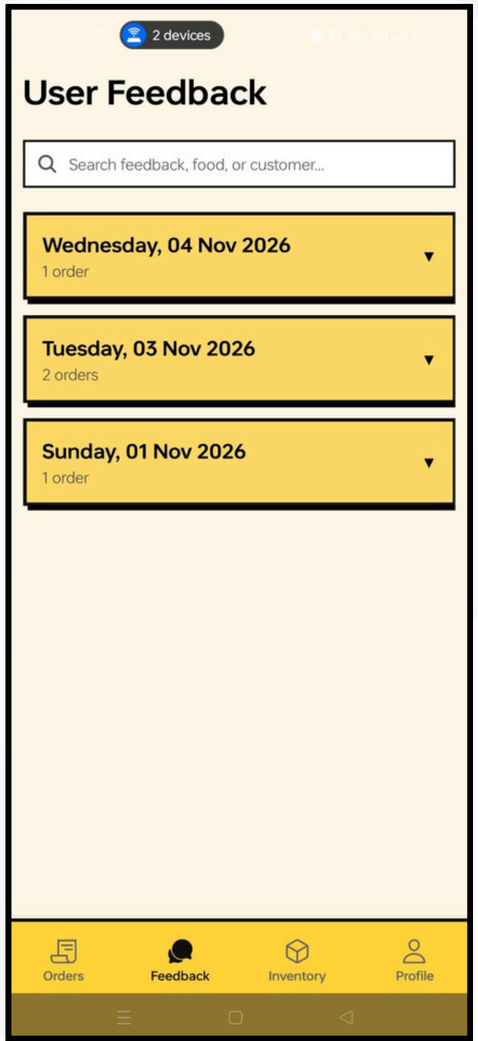
[CUSTOMER]
Orders



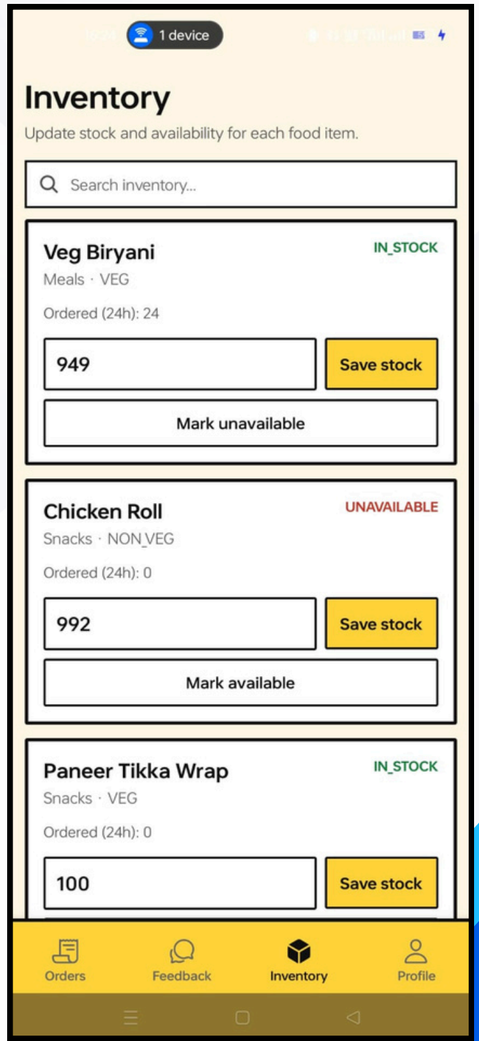
[CUSTOMER]
Provide Feedback



[STAFF]
Review Feedback

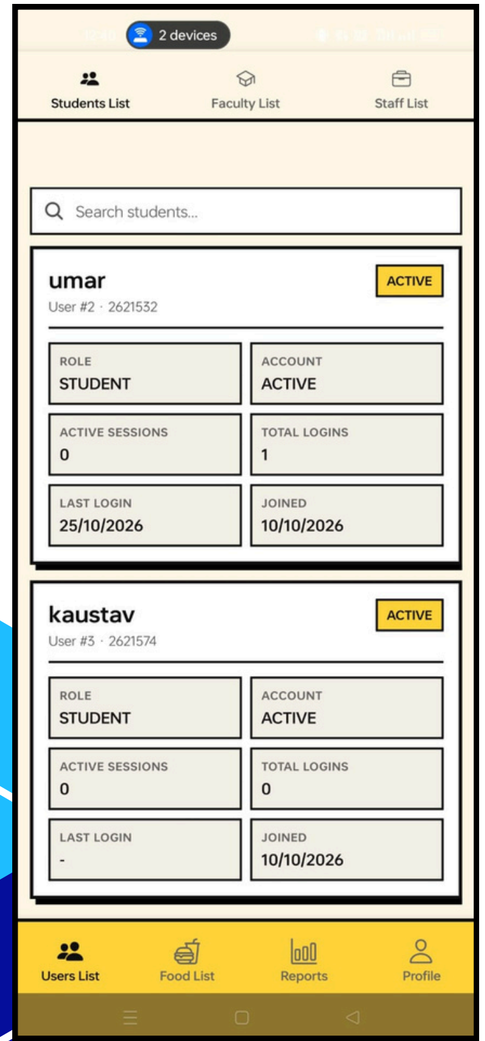


[STAFF]
Update Item Availability

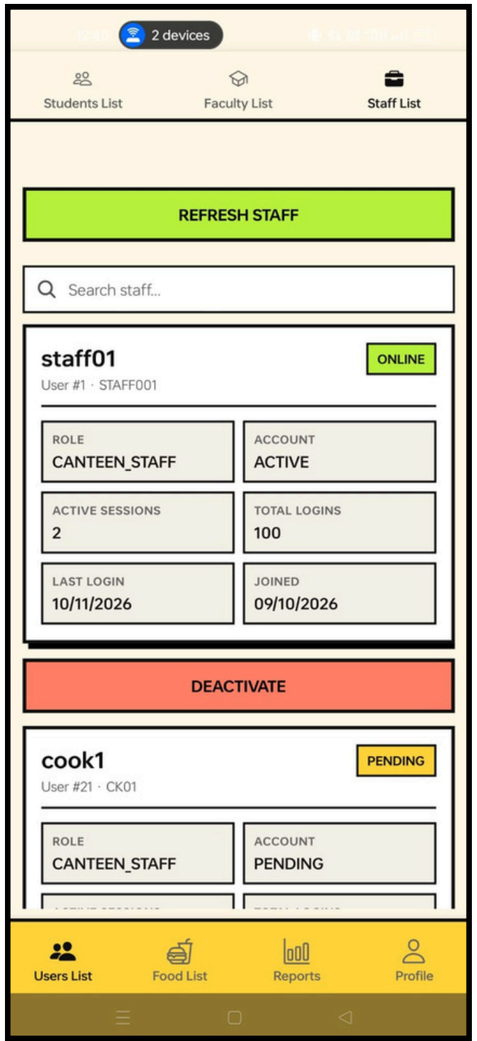


IMPLEMENTATION: Frontend

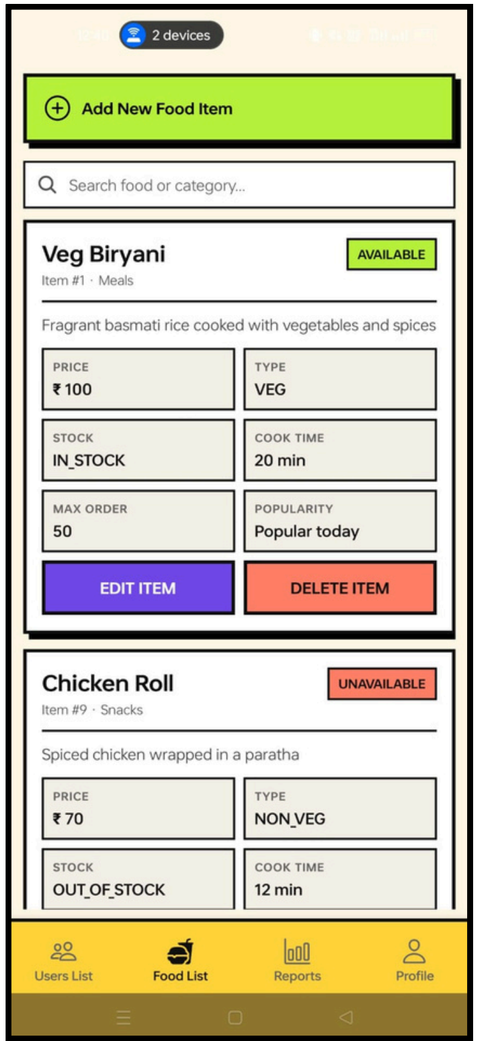
[ADMIN]
Customer List



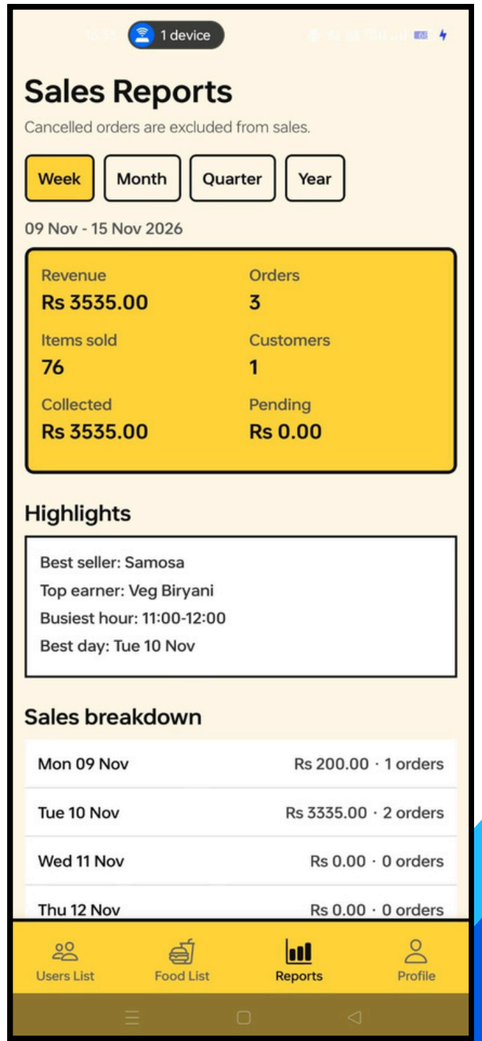
[ADMIN]
Staff List



[ADMIN]
Food Items list



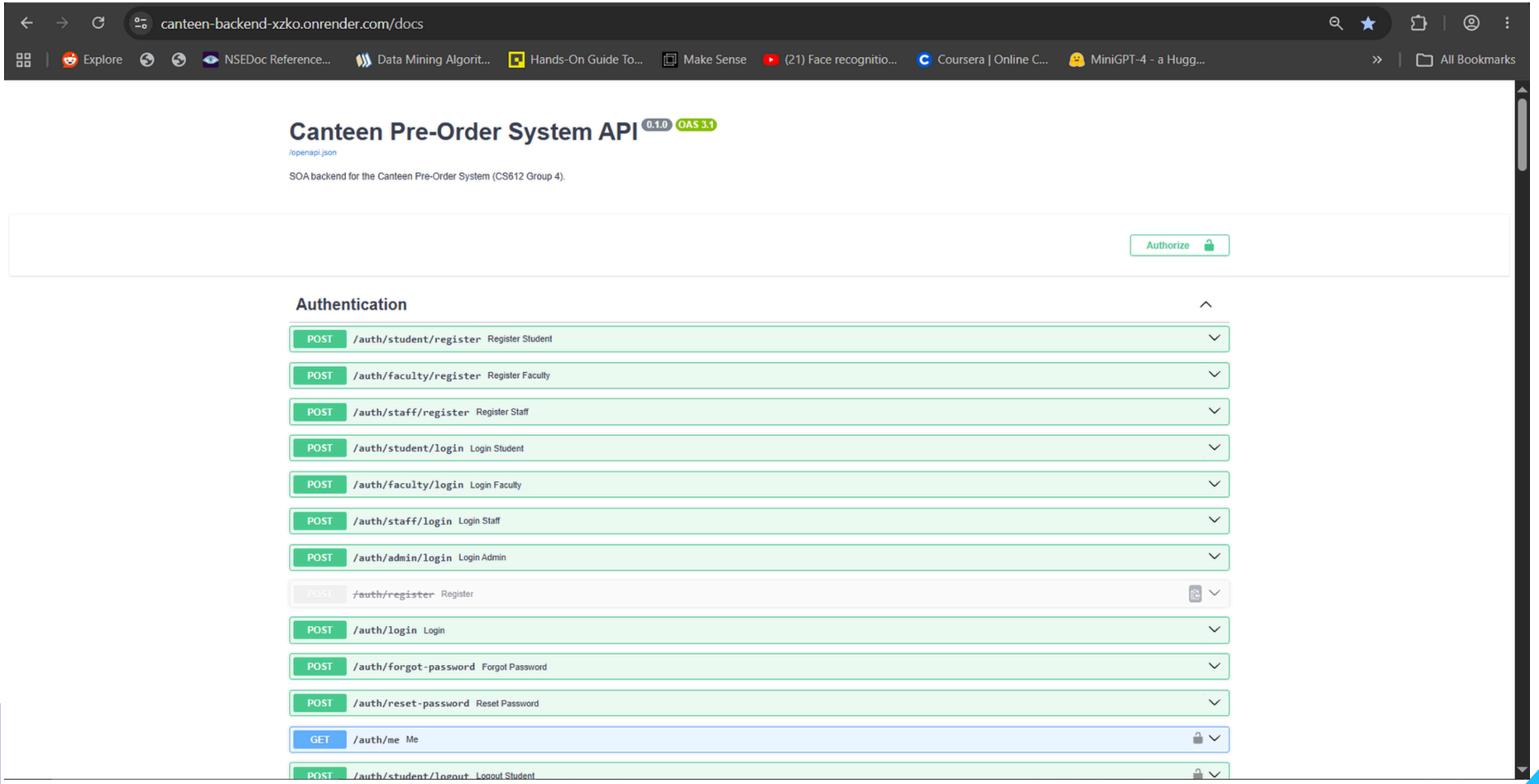
[ADMIN]
Sales Report



DEPLOYMENT

Backend

The backend API is deployed on Render, the base for URL which is:
https://canteen-backend-xzko.onrender.com/



DEPLOYMENT

Frontend

For the frontend, instead of a traditional hosted website, we built a native mobile application and so the users interact with the app via a compiled .apk package, which is configured to send requests directly to our live Render backend endpoints.

LIMITATIONS

- The application has in-app notifications only; push notification feature has not been added yet.
- UPI payments through Razorpay were subject to KYC-related requirements during implementation, which limited the availability of the UPI payment flow for this project. Wallet payments and cash payment at the counter remain supported.

FUTURE SCOPE

- Users will be able to schedule orders in advance for a specific date and time, such as booking lunch in the morning, and the canteen will start preparing the food so that it is ready exactly at the chosen pickup time.

- A group ordering feature will let a class, department or event organiser create one shared order that several people can add items to, with the cost split among them and a single pickup for everyone.

- Prepared food from cancelled orders will be offered at a discounted price through a live surplus-food section, which will reduce food wastage while giving users affordable meals.

- A live canteen rush indicator will show how busy the canteen is at different times of the day, helping users choose quieter pickup times and spreading demand more evenly during peak hours.

CONCLUSION

- The project follows a **Service-Oriented Architecture** in which nine independent services communicate through well-defined REST APIs.
- We developed a fully functional Canteen Pre-Order System that serves four user roles: Students, Faculty, Canteen Staff and the Administrator.
- The system reduces queues and waiting time by allowing users to place orders in advance and collect them with a unique pickup code.
- Customers can browse the menu, add items to the cart, make payment, track their order status in real time and give feedback on their food.
- Canteen staff can process orders through each stage of preparation, update food stock and verify every pickup before the food is handed over.
- The administrator can manage the menu and staff accounts, monitor system activity live and analyse performance through weekly, monthly, quarterly and yearly sales reports.
- Role-based access control, secure password storage and real-time stock checks keep the system secure and prevent items from being oversold.